

Madeline C. Hayes

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EDUCATION:

Master of Science in Remote Sensing and Geospatial Sciences

May 2022

Boston University, Boston, MA

Bachelor of Science in Environmental Sciences

May 2019

University of Vermont, Burlington, VT

Concentration: Environmental Analysis and Assessment | **Minor:** Geospatial Technologies

PROFESSIONAL EXPERIENCE:

Consultant, Geospatial Scientist

November 2024 - Present

Consulting Senior Associate

November 2023 - November 2024

Consulting Associate

June 2022 - November 2023

ERM, Boston MA

- Pioneered and deployed Python-, SQL-, and R-based analytics workflows and geoprocessing tools for large-scale infrastructure and environmental projects across 15+ industries, generating approximately \$250k in annual operational savings through workflow automation and process optimization.
- Developed and fine-tuned a deep learning wetland classification model using transfer learning techniques and limited training data (200 samples, 7 predictor variables), achieving 90% prediction accuracy across a 600,000-acre study area to support infrastructure planning and environmental risk assessment.
- Built a custom Python-based environmental feature crossing analysis tool used extensively in infrastructure routing and siting projects, significantly reducing manual analysis time, minimizing human error, and improving analytical consistency across projects.
- Engineered and maintained enterprise-scale data management systems, leveraging AI-assisted development tools and reusable scripting frameworks to manage 100,000+ environmental features and streamline analytics delivery.
- Published and maintained 100+ client-facing web applications and dashboards using ArcGIS Online and Microsoft Power BI to rapidly prototype analytics solutions and deliver actionable insights under compressed project timelines.

Research Assistant, NASA MEaSUREs

January 2022 - March 2023

Boston University, Center for Remote Sensing

- Collected training and validation data in Google Earth Engine for global annual land cover mapping, achieving an initial model accuracy of 77% through comprehensive satellite imagery analysis, advancing crucial environmental research objectives.
- Conducted in-depth research on land cover information to ensure detailed characterization and accuracy.

Remote Sensing Analyst and Researcher

December 2019 - August 2021

Duke University, Marine Robotics and Remote Sensing Lab

- Developed and deployed a deep learning model for automated seabird detection and enumeration from drone imagery, achieving 92% average accuracy and supporting large-scale ecological monitoring initiatives.
- Created a Python-based predictive model for turbidity forecasting using multispectral drone imagery, reducing manual labor costs by 80% while improving environmental monitoring capabilities.
- Processed and analyzed large spatial and imagery datasets to generate high-resolution 3D models and orthomosaics for environmental analysis and land management applications.

GIS Team Lead and Drone Pilot

June 2018 - November 2019

University of Vermont, Spatial Analysis Lab

- Directed scalable end-to-end data acquisition, processing, and GIS integration workflows, improving operational efficiency, data consistency, and delivery timelines across multidisciplinary projects.
- Developed and integrated high-resolution geospatial datasets and analytical layers to enhance statewide land cover modeling capabilities, producing a dataset approximately 900 times more detailed than previous versions and significantly improving spatial accuracy.
- Trained, mentored, and coordinated a team of 20 technicians and drone pilots, establishing standardized workflows, QA/QC procedures, and operational best practices.

RESEARCH EXPERIENCE:

Remote Sensing for Water Quality Research

July 2020 - July 2021

Duke University, Marine Robotics and Remote Sensing Lab

- Collaborated with the North Carolina Department of Environmental Quality to determine the utility of drone technology for water quality assessment
- Conducted multispectral drone flight operations before, during, and after dredging events
- Created a Python programming workflow to extract turbidity measurements from raw images, including radiometric calibration and georeferencing of imagery
- Generated turbidity heat maps and image mosaics for comparison to in-situ measurements

Artificial Intelligence for Seabird Population Monitoring Research

January 2020 - May 2021

Duke University, Marine Robotics and Remote Sensing Lab

- Processed drone imagery collected by the Wildlife Conservation Society in Argentina to generate orthorectified maps with photogrammetric techniques
- Built, trained, validated, and deployed a convolutional neural network for the automated detection and enumeration of seabirds in the Falkland Islands, Argentina
- Ran inference on entire seabird colony areas to generate automated detections
- Evaluated geostatistical techniques for breeding pair and active nest analysis

PRESENTATIONS

Hayes, M.C. (April 2021). "Deep learning and drones to automate seabird population counts," ESRI Imagery and Remote Sensing Educators Summit.

Hayes, M.C. (March 2021). "Deep learning and drones to automate seabird population counts," Drones in the Coastal Zone Workshop.

Hayes, M.C. (November 2019). "Mapping water chestnut from above," North American Lake Management Society Symposium, Burlington, VT.

Hayes, M.C. (April 2019). "Mapping water chestnut from above," University of Vermont Student Research Conference, Burlington, VT.

PAPERS

Hayes, M.C., P.C. Gray, G. Harris, W.C. Sedgwick, V.D. Crawford, N. Chazal, S. Crofts, and D.W. Johnston (2021). Drones and deep learning produce accurate and efficient monitoring of large-scale seabird colonies.

Ornithological Applications, 123, 1-16. DOI: 10.1093/ornithapp/duab022

Hayes, M.C., B. Puckett, C. Deaton, J.T. Ridge (2021). Estimating dredge-induced turbidity using drone imagery.

Preprints, 2022010424. DOI: 10.20944/preprints202201.0424.v1

TEACHING EXPERIENCE:

Teaching Fellow, Crises of Planet Earth

January 2022 - May 2022

Boston University, College of Arts and Sciences

Teaching Fellow, Introduction to Climate and Earth System Science

September 2021 - December 2021

Boston University, College of Arts and Sciences

Teaching Assistant, Intro to GIS

August 2018 - December 2018

University of Vermont, Rubenstein School of Environment and Natural Resources

SKILLS:

Programming and Analytics: Python (NumPy, PyTorch, TensorFlow, OpenCV, scikit-learn, ArcPy)

Machine Learning and Modeling: Predictive Modeling, Deep Learning, Computer Vision, Classification Models, Spatial Analytics

Data Visualization and Reporting: Power BI, Plotly, Matplotlib, ArcGIS Dashboards

Data Governance and Automation: ETL workflows, SQL databases, Git, AI-assisted development workflows

Geospatial Technologies: ESRI (ArcGIS Pro, ArcGIS Online, ArcGIS Server, ArcSDE), Remote Sensing, LiDAR Analysis